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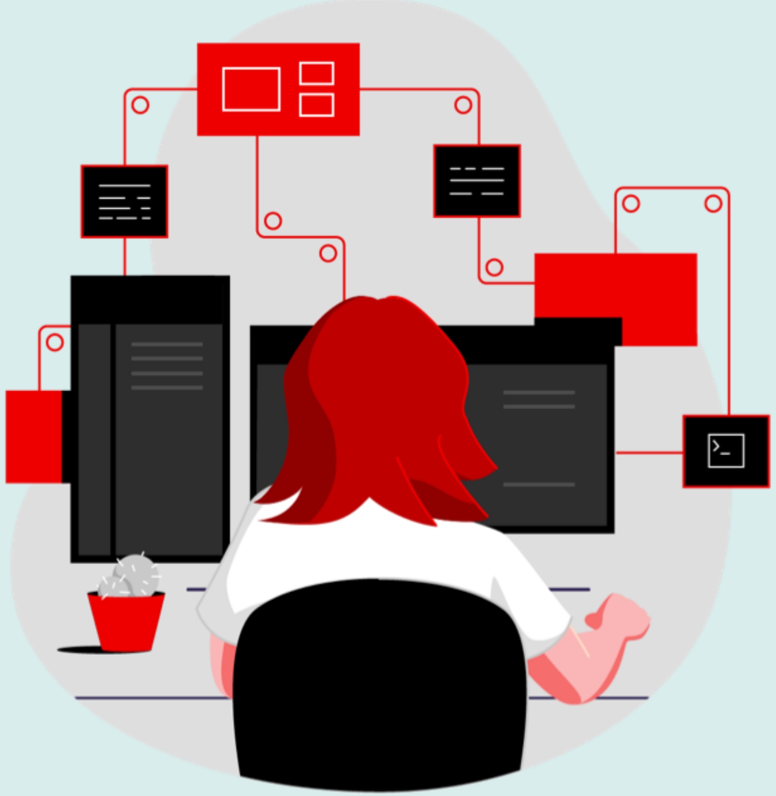
Automated Control System Based on Simatic S7-1200

FINAL EXAMINATION PROJECT



AUTOMATED FINAL ASSEMBLY & PACKAGING SYSTEM

Using Siemens S7-1200 PLC and HMI, Factory IO

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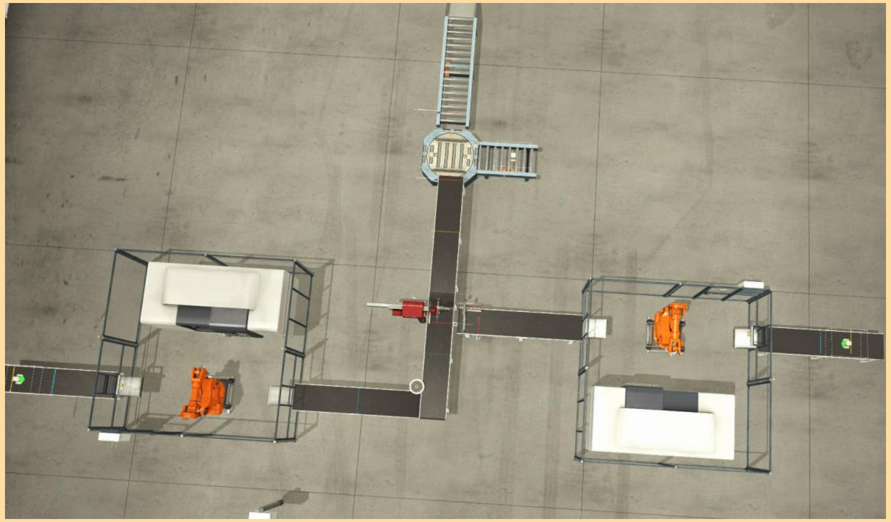
INTRODUCTION

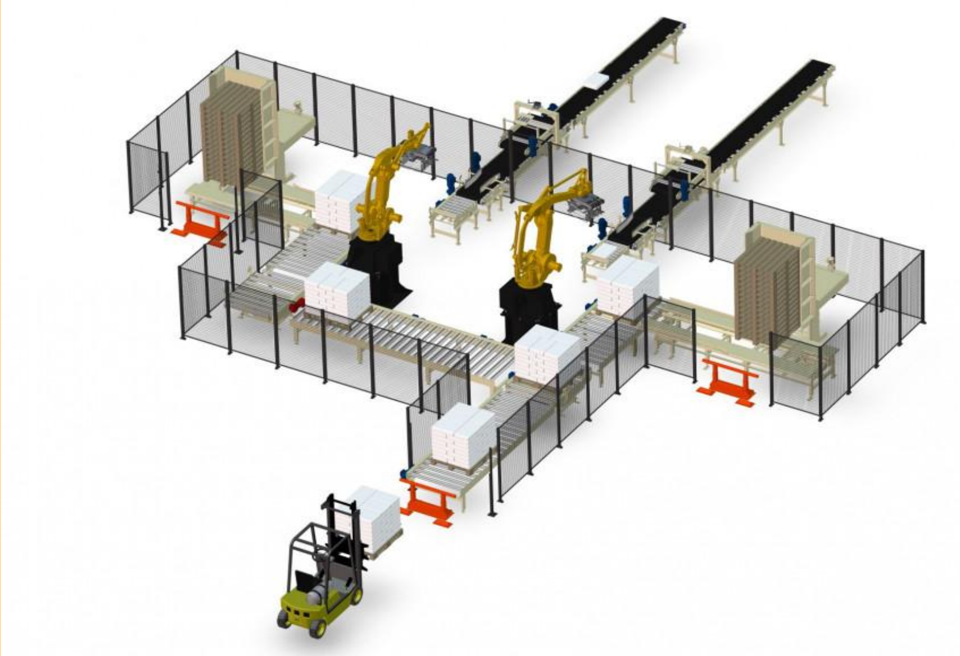
The main objectives of this project are:

- **To design a PLC-based control system for a dual-stream automated assembly and packaging process [TIA PORTAL V19]**
- **To develop structured and reliable PLC logic for conveyors, robotic handling, and machining centers**
- **To design an intuitive HMI for operator monitoring and control**
- **To simulate and test the system under automatic and manual operating modes**
- **To demonstrate safe, coordinated, and efficient system operation (for simulation used Factory IO)**

PROJECT SUMMARY

Industrial automation improves productivity, accuracy, and safety in modern manufacturing. PLC and HMI systems are widely used to control complex industrial processes.
This project presents an **Automated Final Module Assembly & Packaging System** with a **dual-stream workflow**. Two parallel conveyors feed modules to robotic arms, which load them into Machining Centers for processing. After processing, the modules are assembled into one unit and transferred to a packaging station for sealing and final output.





PLC I/O LIST

A comprehensive list of all input and output tags used in the system's control logic is provided below. Inputs include sensor feedback (e.g., Base_Detector, Lid_Aligned), while outputs control actuators such as clamps, conveyors, and robotic movements (Gripper_Move X, Product_Output_Conveyor). All tags are mapped to standard Boolean variables with assigned memory addresses for seamless integration with TIA Portal and Factory IO simulation.

«Start» , «Stop» , «E-STOP» ➔ HMI Buttons
«Base_Detector», «Lid_Detector» ➔ Sensors
«Gripper_Moving_X/Z» ➔ Cartesian robot signals

You can view the complete PLC tags and all mechanisms in the report

Input

PLC tags

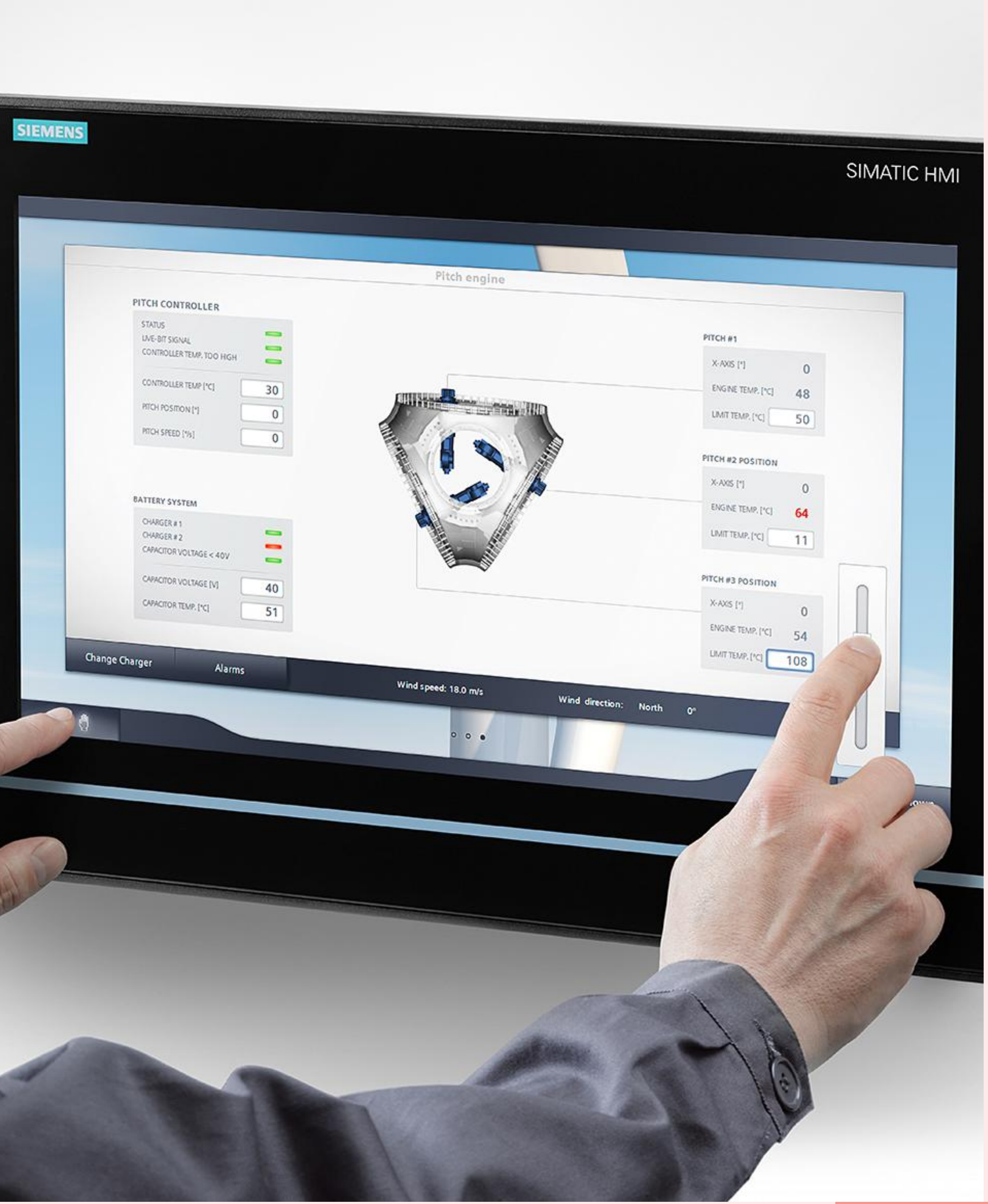
	Name	Tag table	Data type	Address	Retain	Acces...	Writa...	Visibl...	Comment
1	Start	Standard-Varia...	Bool	%I0.0	☐	☑	☑	☑	
2	Stop	Standard-Variable...	Bool	%I0.1	☐	☑	☑	☑	
3	Base_Detector	Standard-Variable...	Bool	%I0.2	☐	☑	☑	☑	
4	Base Clamped	Standard-Variable...	Bool	%I0.3	☐	☑	☑	☑	
5	Base_Clamp_Raised	Standard-Variable...	Bool	%I0.4	☐	☑	☑	☑	
6	Lid_Detector	Standard-Variable...	Bool	%I0.5	☐	☑	☑	☑	
7	Lid Clamped	Standard-Variable...	Bool	%I0.6	☐	☑	☑	☑	
8	Gripper_Moving_X	Standard-Variable...	Bool	%I0.7	☐	☑	☑	☑	
9	Gripper Moving Z	Standard-Variable...	Bool	%I1.0	☐	☑	☑	☑	

Output

	Name	Tag table	Data type	Address	Retain	Acces...	Writa...	Visibl...
16	Base_Material_emitter	Standard-Variable...	Bool	%Q3.0	☐	☑	☑	☑
17	Lid_Material_Emitter	Standard-Variable...	Bool	%Q0.1	☐	☑	☑	☑
18	Base_Clamp	Standard-Variable...	Bool	%Q0.2	☐	☑	☑	☑
19	Lid_Output_Conveyor	Standard-Variable...	Bool	%Q0.3	☐	☑	☑	☑
20	base Clamp Raise	Standard-Variable...	Bool	%Q0.4	☐	☑	☑	☑
21	Lid_Clamp	Standard-Variable...	Bool	%Q0.5	☐	☑	☑	☑
22	Bottle Output Conveyor	Standard-Variable...	Bool	%Q0.6	☐	☑	☑	☑
23	Product_Output_Conveyor	Standard-Variable...	Bool	%Q0.7	☐	☑	☑	☑
24	Collection_Box_emitter	Standard-Variable...	Bool	%Q1.0	☐	☑	☑	☑
25	Gripper-Move X	Standard-Variable...	Bool	%Q1.1	☐	☑	☑	☑
26	Gripper-Move Z	Standard-Variable...	Bool	%Q1.2	☐	☑	☑	☑

Memory

	Name	Tag table	Data type	Address	Retain	Acces...	Writa...	Visibl...
31	Base Aligned	Standard-Variable...	Bool	%M0.1	☐	☑	☑	☑
32	Lid Aligned	Standard-Variable...	Bool	%M0.2	☐	☑	☑	☑
33	Lid Grabbed	Standard-Variable...	Bool	%M0.3	☐	☑	☑	☑
34	Lid Above 'Lid Clamp'	Standard-Variable...	Bool	%M0.4	☐	☑	☑	☑
35	Lid Above Base	Standard-Variable...	Bool	%M0.5	☐	☑	☑	☑
36	Product Assembled	Standard-Variable...	Bool	%M0.6	☐	☑	☑	☑
37	Emitter	Standard-Variable...	Bool	%M0.7	☐	☑	☑	☑



COMPONENTS USED

TON (On-Delay Timer)

•**Definition:** Timer that starts counting when the input turns ON; output becomes TRUE only after preset time elapses.

•**Function:** Delays turning ON the output for a defined time after the input is activated.

TOF (Off-Delay Timer)

•**Definition:** Timer that starts counting when the input turns OFF; output remains TRUE during counting, then turns OFF after preset time.

•**Function:** Keeps the output ON for a defined time after the input goes OFF.

TP (Pulse Timer / One-Shot Timer)

•**Definition:** Timer that generates a fixed-duration pulse when the input turns ON.

•**Function:** Produces a single timed pulse regardless of input duration.

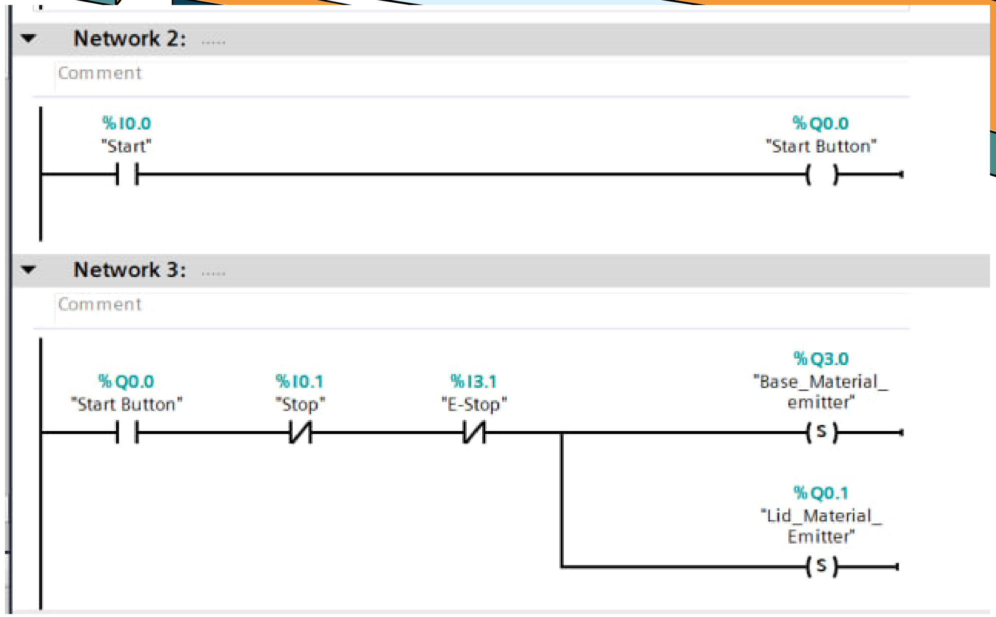
CTU (Count-Up Counter)

•**Definition:** Counter that increments its value by 1 each time the input signal transitions from FALSE to TRUE.

•**Function:** Counts events or pulses up to a preset value.

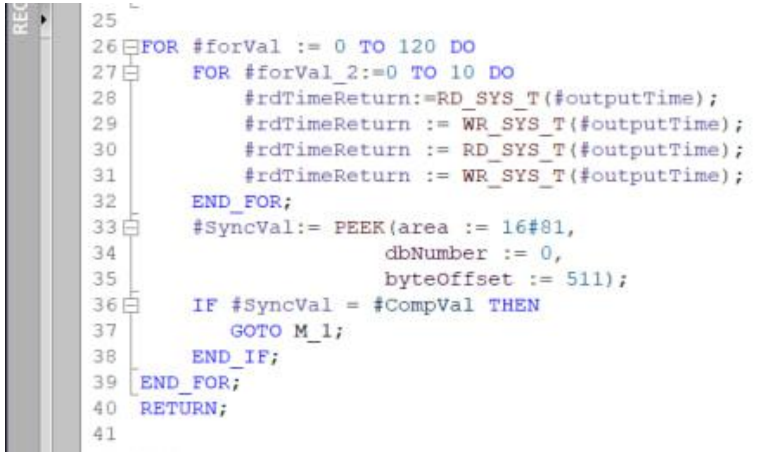
PLC LOGIC (LAD AND SCL)

- In TIA Portal, we used the LAD language in the Main Block. The main logic is implemented here. On the right side, you can see a small part of the PLC logic.
- In the Function Block, we wrote the logic using SCL. A small part of this logic can be seen in the image on the right



Operating Modes

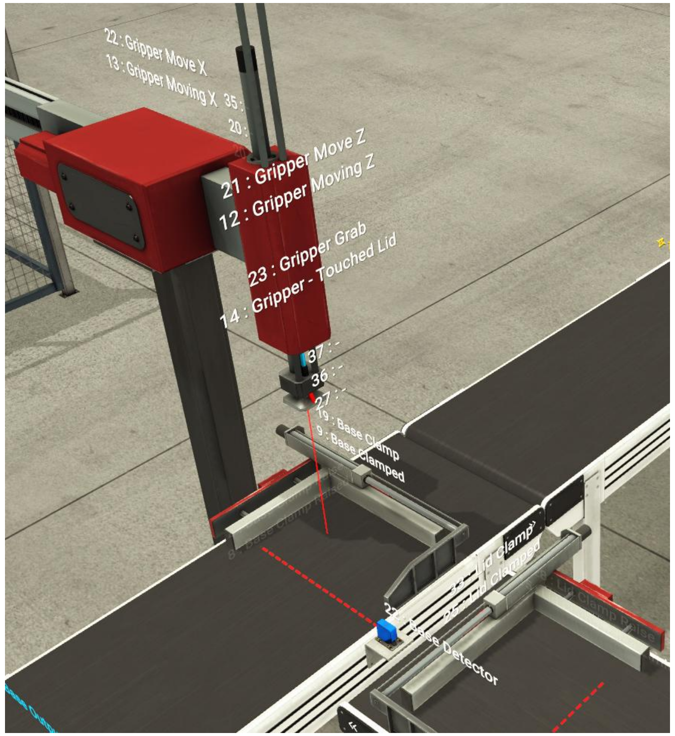
- **Manual Mode:**
  Operators can manually control conveyors, robotic arms, and machining centers. This mode is mainly used for system testing, setup, and troubleshooting.
- **Automatic Mode:**
  The system runs fully automatically according to a predefined sequence. Transitions between process steps are based on sensor feedback, timers, and interlock conditions.



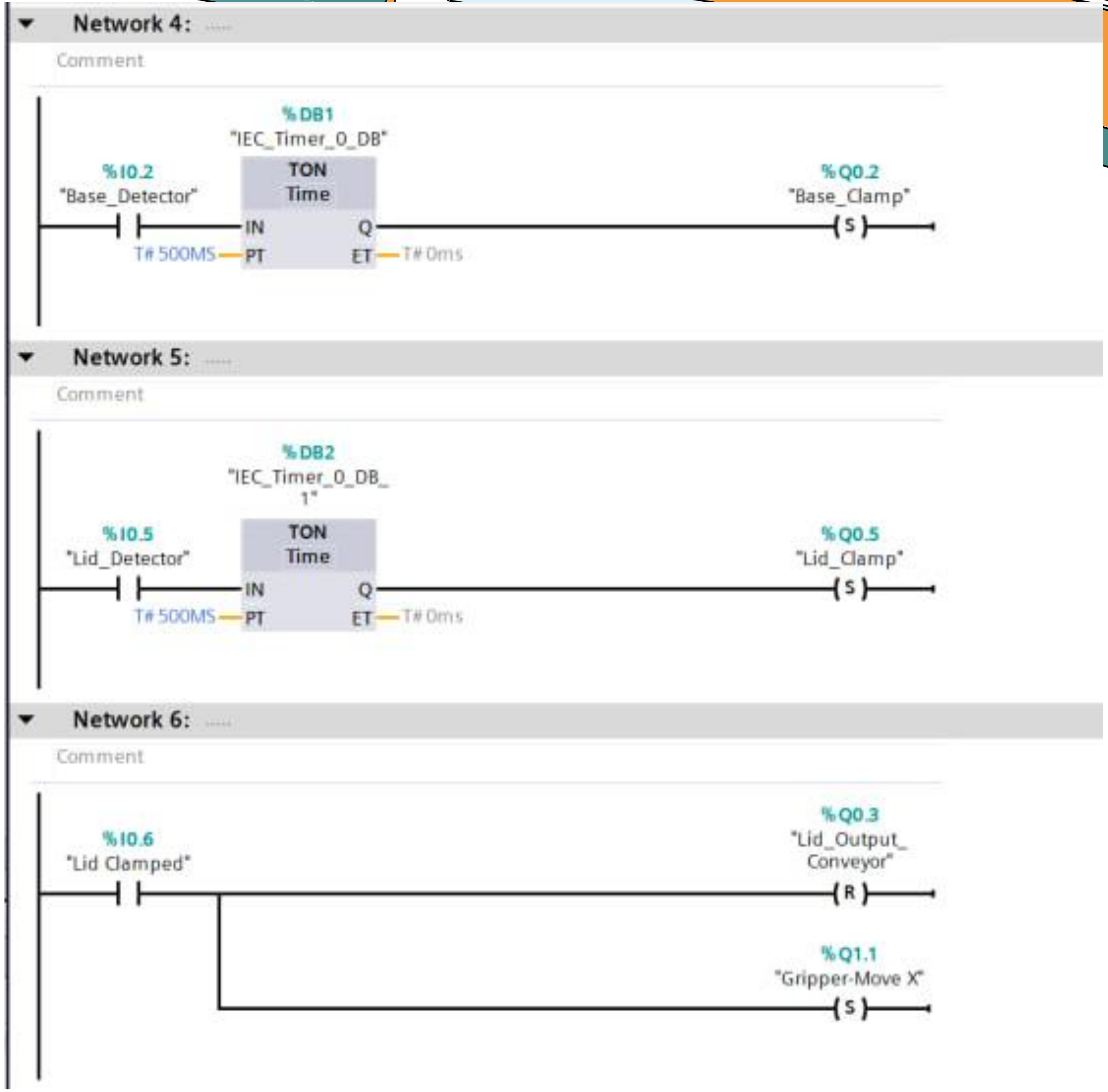
```
FOR #forVal := 0 TO 120 DO
    FOR #forVal_2:=0 TO 10 DO
        #rdTimeReturn:=RD_SYS_T(#outputTime);
        #rdTimeReturn := WR_SYS_T(#outputTime);
        #rdTimeReturn := RD_SYS_T(#outputTime);
        #rdTimeReturn := WR_SYS_T(#outputTime);
    END_FOR;
    #SyncVal:= PEEK(area := 16#81,
                    dbNumber := 0,
                    byteOffset := 511);
    IF #SyncVal = #CompVal THEN
      GOTO M_1;
    END_IF;
END_FOR;
RETURN;
```

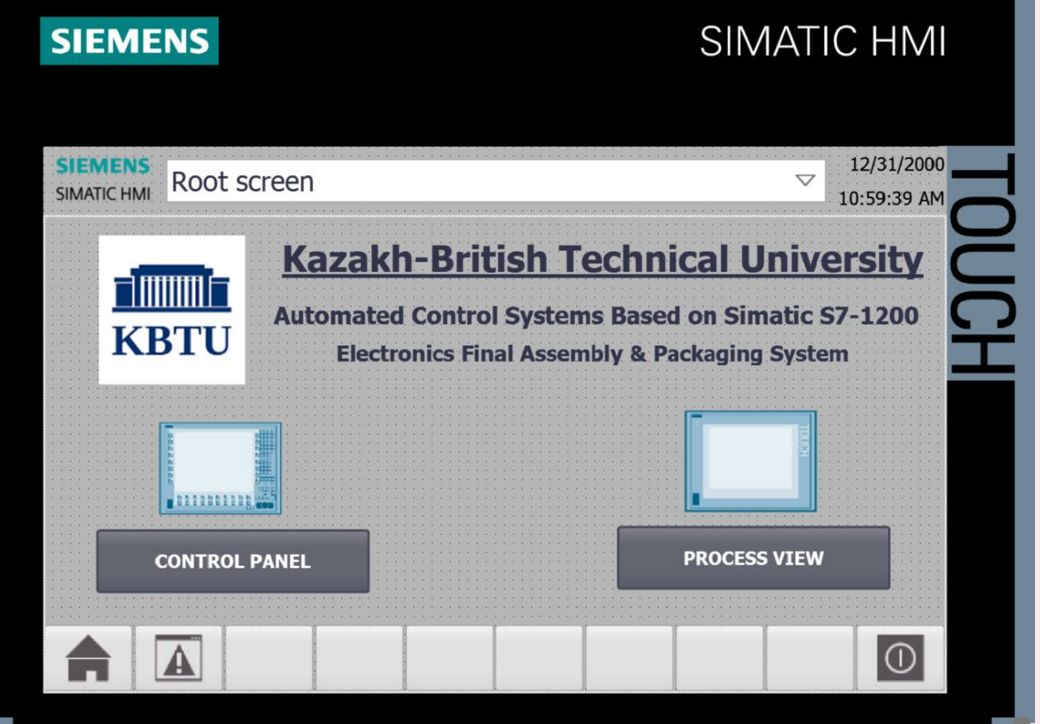
TIA Portal

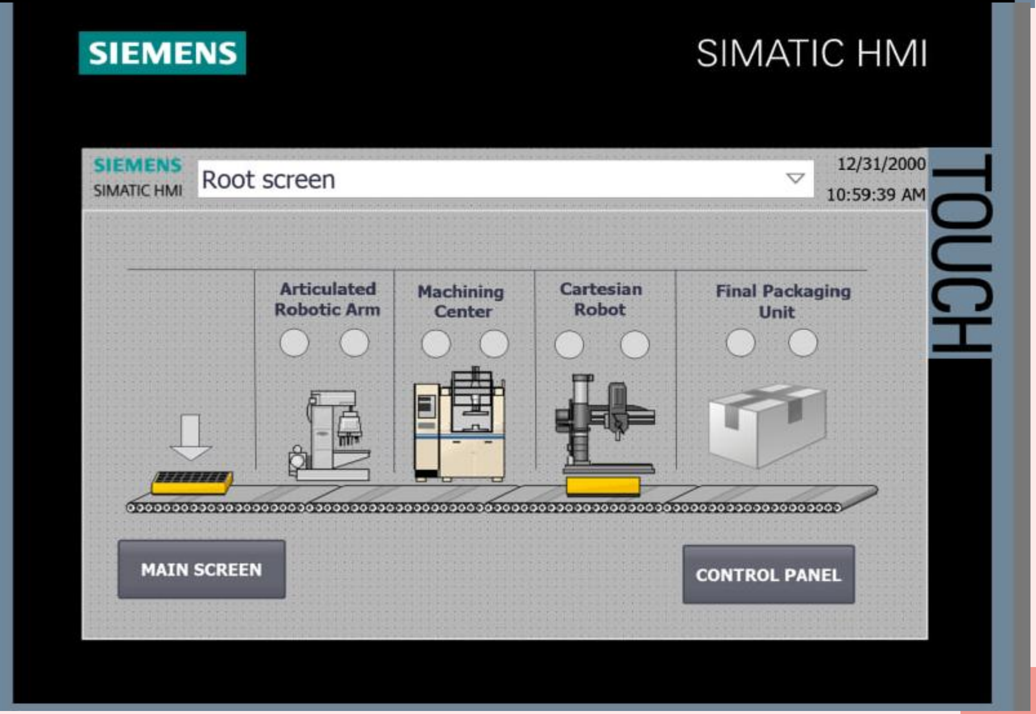
In this slide, we will explain only the most important parts of the PLC programming.

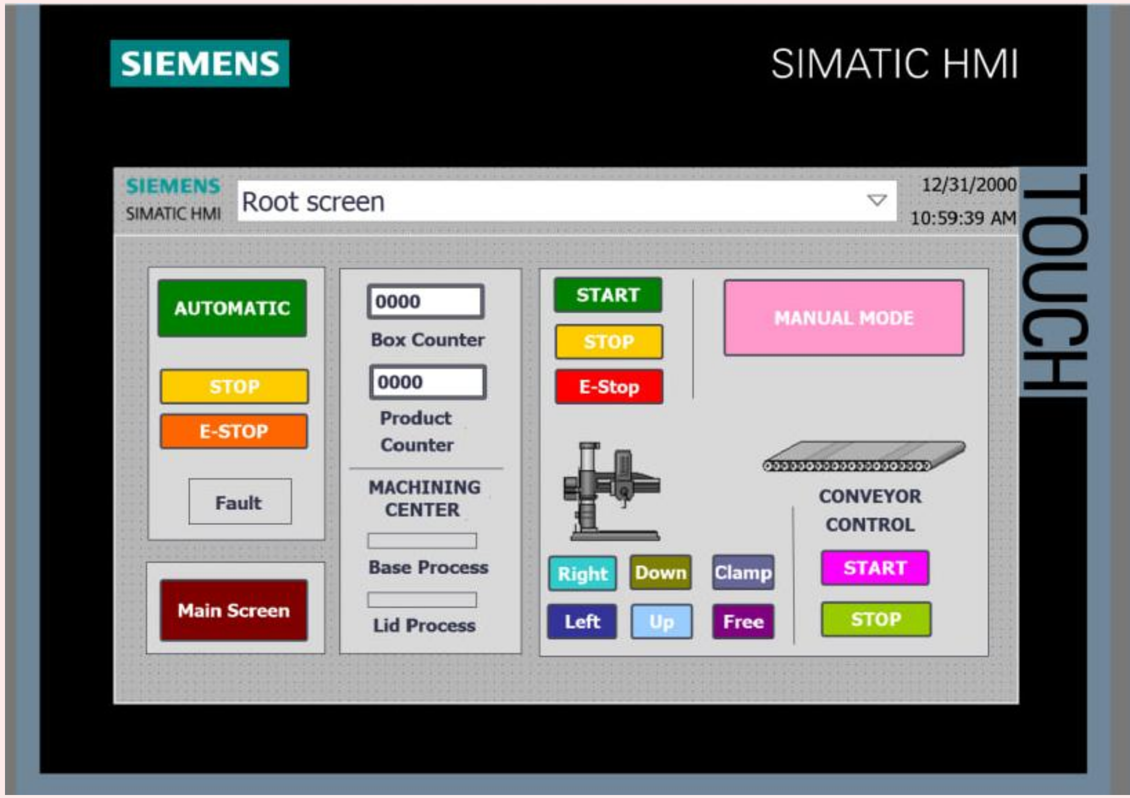


The complete PLC programming logic can be found in the report.









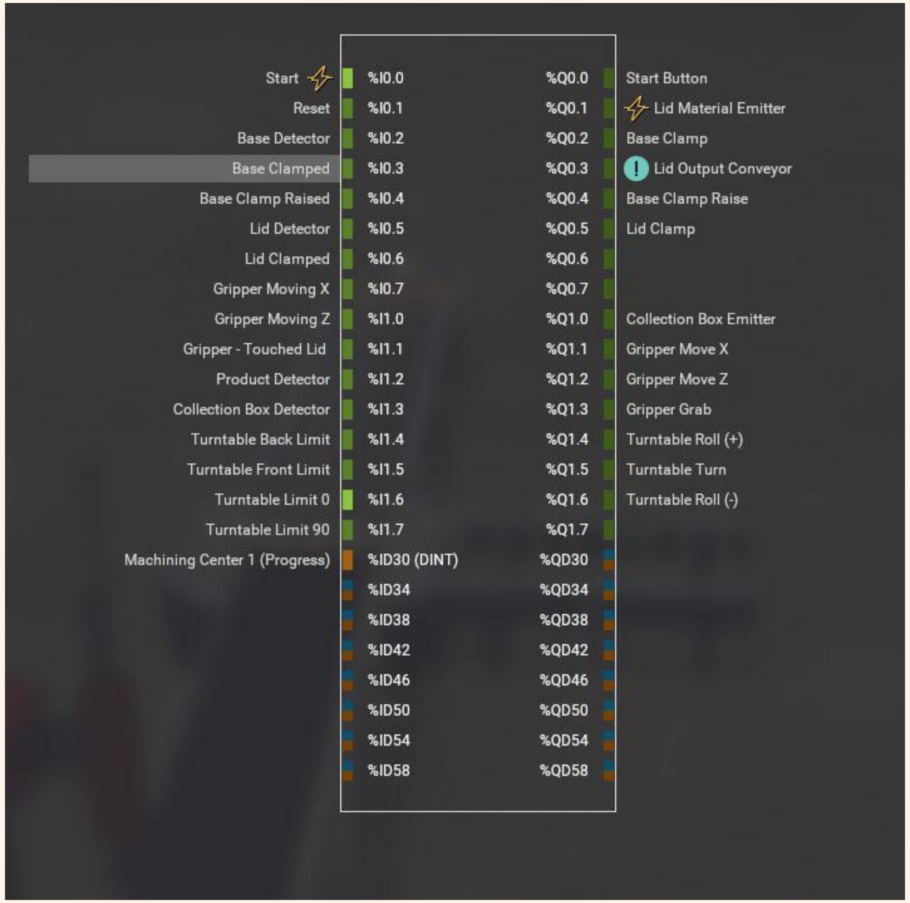
Main Screen

Control Panel

Process VIEW

HMI SCREEN DESIGN

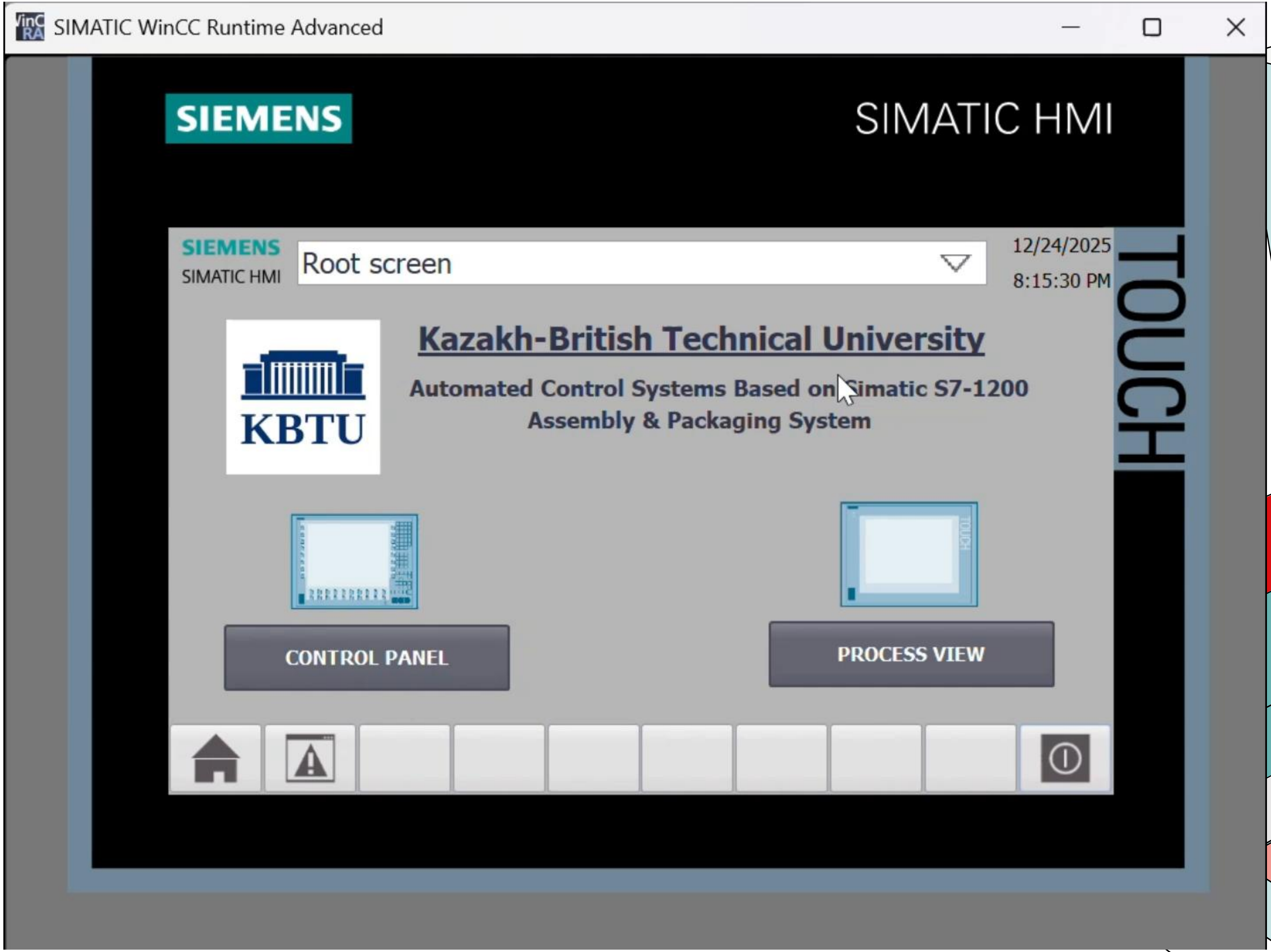
«FACTORY IO» FOR SIMULATION



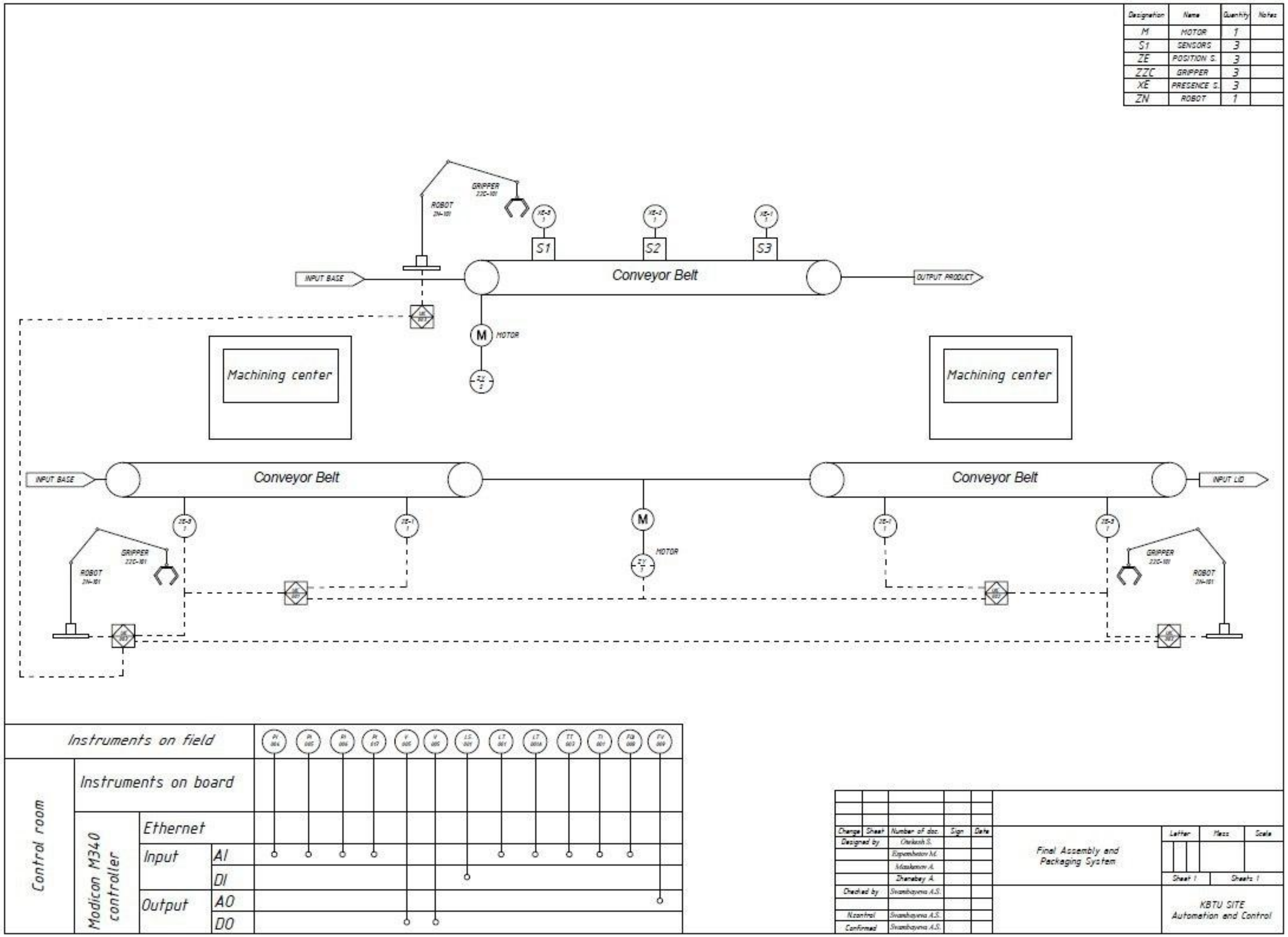
Factory I/O software was used to simulate and test the automated assembly and packaging system. Factory I/O provides a realistic 3D industrial environment that allows verification of PLC control logic, sensor feedback, and actuator behavior without the need for physical hardware.

The PLC program developed in TIA Portal was connected to Factory I/O via industrial communication. All system components, including conveyors, robotic arms, machining centers, assembly stations, and the packaging unit, were mapped to their corresponding PLC inputs and outputs. Simulation tests were performed in both manual and automatic operating modes. In manual mode, individual actuators were tested to verify correct I/O addressing and basic functionality. In automatic mode, the complete process sequence was executed, starting from module entry on the conveyors to final packaging and output.

VIDEO DEMONS TRATION



AUTOCAD SCHEME



RESULTS & PERFORMANCE

The automated final module assembly and packaging system was successfully tested in Factory I/O using TIA Portal PLC logic. The system performed correctly in both manual and automatic modes, demonstrating smooth and coordinated operation of conveyors, robotic arms, machining centers, assembly, and packaging stations.

- **Correct Sequencing:**
- Modules moved through the process in the correct order
- Robotic arms and machining centers operated without collisions
- **HMI Functionality:**
- Operators could start/stop the system, switch modes, and monitor status
- Alarm indicators and feedback signals worked as intended
- **Safety Interlocks:**
- Conveyors and robots did not operate when safety conditions were violated
- Emergency stop function immediately stopped all operations

RESULT

CONCLUSION

•The system successfully demonstrates **dual-stream automation** for assembly and packaging.
•PLC logic and HMI design ensure **safe, efficient, and reliable operation**.
•Factory I/O simulation validates the **full process flow** without physical hardware.
•The project confirms that **automation improves throughput and reduces human error**.

The dual-stream automated assembly and packaging system works correctly and safely.
PLC logic and HMI ensure efficient operation and accurate process control.
Simulation in Factory I/O confirms reliable sequencing and improved throughput.